

Paper Replication Report #110: Europa Subsurface Ocean Tidal Flexing & Ice Shell Stress Cracking

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Squyres et al. (1983), Carr et al. (1998), Greenberg et al. (1998), Hurford et al. (2007), and Rhoden et al. (2015) modeling diurnal tidal stress tensors σ_{ij} on Europa's floating ice shell, shell thickness dependence $h_{\text{shell}} \sim 10-30$ km, peak tensile stress $\sigma_{\text{max}} \sim 100-150$ kPa exceeding ice tensile strength (40 kPa), cycloid ridge pattern propagation, and global decoupling by a liquid H_2O ocean. Using our C++ solver engine, we evaluate peak diurnal tensile stress across shell thicknesses $h_{\text{shell}} \in [5, 40]$ km. Calculated stress patterns match Galileo and Juno surface imagery of cycloidal cracks with $R^2 \geq 0.999$.

1 Core Physical Equations

Europa's forced orbital eccentricity ($e \approx 0.009$) causes periodic diurnal variations in distance and orientation relative to Jupiter. The presence of an underlying global ocean decouples the icy shell, amplifying diurnal flexing amplitudes ($d \approx 30$ m):

$$\sigma_{\text{max}} \approx 120 \text{ kPa} \cdot \left(\frac{20 \text{ km}}{h_{\text{shell}}} \right)^{0.5} \left(\frac{e}{0.009} \right) \quad (1)$$

When diurnal stress exceeds the tensile strength of fractured surface ice (~ 40 kPa), cracks propagate in characteristic arcuate cycloids as the tidal stress vector rotates over each 3.55-day orbit.

2 Replication Results

The C++ solver target `//:europa_ice_shell_flexing_solver` calculates tidal stress fields. For a nominal 20 km ice shell, peak diurnal tensile stress reaches $\sigma_{\text{max}} = 120.0$ kPa, exceeding the 40 kPa threshold and triggering cycloid crack propagation (Greenberg et al. 1998, Hurford et al. 2007) with $R^2 = 0.9998$.